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Manufacturing

“SIMULATION TOOLS Can SAVE Up To 70 Per Cent Of The MANUFACTURING COST”

Large manufacturing businesses are readily taking up additive manufacturing technologies and investing big in simulation tools that optimise manufacturing. When will SMEs be able to afford these? And where does the business opportunity lie in the Indian manufacturing scenario? Sumanth Kumar, vice president, SIMULIA Growth, Dassault Systemes, shares his views with Paromik Chakraborty of Electronics For You

Q. How big is the presence of additive manufacturing in Indian industries?

A. The reach of additive manufacturing has blossomed in the last two-three years. In additive manufacturing, there are two focus areas for manufacturers: printing polymer or plastics and printing metals. Several options of 3D printer systems are available for both of these categories.

The ecosystem has seen inclusion of new companies that print plastics and metals almost on a monthly basis. Many of the hardware printers produced by European and American companies that have global presence are being leveraged by manufacturers. Aerospace and defence industry is leading from the forefront in adopting additive manufacturing. Automotive industry is not far behind. Life sciences industry has also taken a step forward.

With the rising popularity of additive manufacturing, digital simulation tools are finding great demand. Top manufacturers in India are showing interest in simulation tools. General Electric is one of them. The divisions of Boeing and Airbus in India are also keen on taking up these solutions.

Q. How do simulation tools improve cost- and quality-control in additive manufacturing?

A. Cost savings come from reduction in material being used. This,

in turn, reduces throwaways and helps users to get the right prototype printed at the first go, saving a lot of energy and effort. From the feedback of our customers, on an average, digital simulation tools like Simulia enable up to 30-35 per cent weight savings in materials used. So if you have been doing additive manufacturing without using digital tools, you will be able to save 60-70 per cent costs by investing in them.

Q. How big is the business opportunity for India in the additive manufacturing domain?

A. Additive manufacturing is an emerging technology. A year ago, there were eight different workflows for additive manufacturing. Today, this number has increased to 14. Every single method has its own process of printing. So businesses have to buy printers, material and hardware relevant to that method. They do not know which method will survive through time and which all may die out.

Keeping that context in mind, not only for India but on a global scale, these services companies can play a big role by studying and trying out these methods and providing consultancy to manufacturers. The biggest challenge in additive manufacturing is transitioning from the prototyping phase to the industrial

volume production phase. That's where services companies can play a major role in providing the right suggestions to manufacturers.

Q. What about simulation tools for additive manufacturing?

A. While searching for a business opportunity, the important point of consideration should be the barriers in the selected field and the upscale investment needed. The barrier in simulation technology development is very high and entry is expensive. We are using technology which we have been developing and working on for the last 35 years. While on the services side the cost is lower.

Q. Cost-wise, how attractive is it for SMEs?

A. We are talking about massive printing systems, which require a big investment. Right now, mostly large companies in the industry are using this technology. The challenge is how to make this technology affordable for SMEs.

Q. How does Simulia fit into additive manufacturing as a digital solution and what role does it play?

A. Simulia is a digital model of 3D printing. It simulates the manufacturing processes virtually and predicts the different behaviours and challenges. **EPF**